



Specifying and Analyzing Transactional Consistency Models with Predicates

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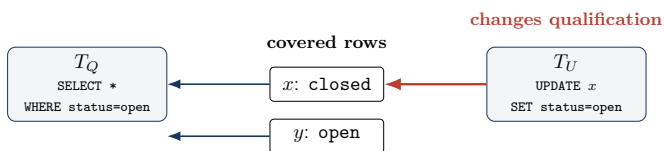
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A semantic and graph-theoretic foundation for predicate operations

A predicate result contains more than returned rows



result: $\{y\}$

Key observation: $x \notin \text{result}$
still records an observation of x .

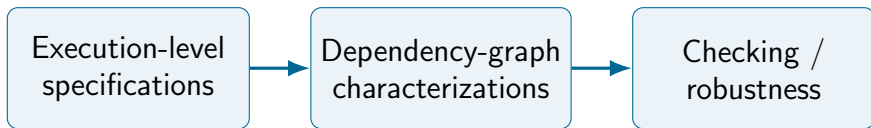
Item reads: value observed

Predicate reads: returned rows and excluded rows

Phantom-style interactions need an explanation for both.

Berenson et al., 1995; Adya, 1999

The gap: familiar foundations stop at named items



Predicate operations are missing from the semantic-graph correspondence.

Question: Can we recover an exact, predicate-aware foundation for SER and SI?

Semantic anchor: explain every covered item

VIS/AR execution framework

Visible transactions

provide the version set



Latest visible writer

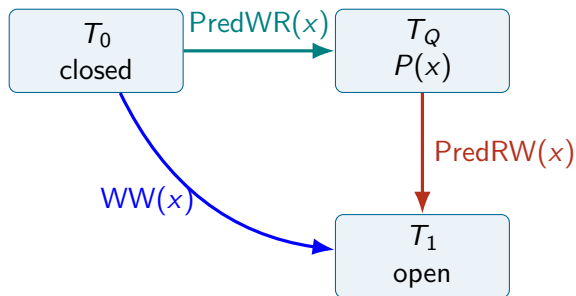
explains each covered item

Returned item: witness matches P

Omitted item: witness does not match P

Cerone, Bernardi, and Gotsman, 2015

From a semantic witness to graph edges



PredWR: version witness

PredRW: later writer changes the
observed result

Result-sensitive: $\Delta(T_0, T_Q, T_1, x)$

The graph records why the result was observed — including the absence of x .

Exact graph characterizations

Serializability

$$\mathcal{H} \models \text{SER}$$



internally consistent +
acyclic predicate-aware graph

Snapshot isolation

$$\mathcal{H} \models \text{SI}$$



internally consistent +
every cycle has **two adjacent**
RW / PredRW edges

Not analogy: both equivalences are proved against the predicate-aware axioms.

Cerone and Gotsman, 2018

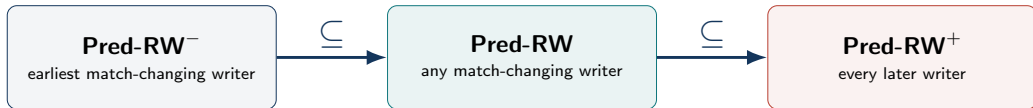
Why the proof bridge matters



observable histories $\longleftrightarrow$ executions

- Predicate graphs are **observable-history** objects.
- Axiomatics supplies the **semantic** correctness target.
- The bridge validates graph-based analysis beyond item-only workloads.

Related work disagrees on one crucial edge



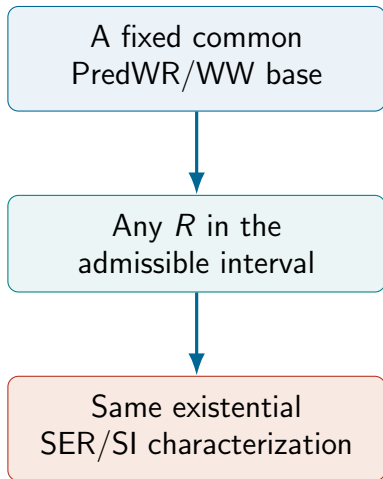
Same existential history-level characterization of SER and SI
edge sets and checking cost may differ

Sun and Zou, 2025

Adya, 1999; “Generalized Isolation
Level Definitions” 2000

Bouajjani, Enea, and Román-Calvo,
2025

What the design space tells us



The theorem is semantic, not a claim that edge sets coincide.

Practical consequence: choose an admissible edge policy for an analysis, while retaining its history-level meaning.

Checking cost can still differ.

Future work: turn the foundation into analysis tools

Efficient black-box
checking

Recover version sets
and version order

Robustness analysis

Extend static reasoning
to predicate operations

Both need a semantics-preserving predicate dependency graph.

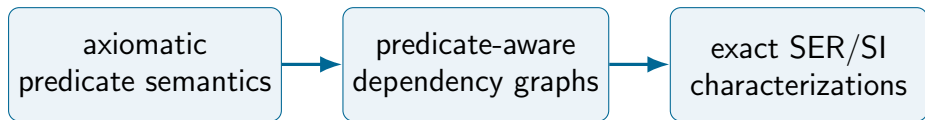
References I

-  Adya, A. (1999). “Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions”. PhD thesis. USA.
-  Berenson, Hal, Phil Bernstein, Jim Gray, Jim Melton, Elizabeth O’Neil, and Patrick O’Neil (1995). “A critique of ANSI SQL isolation levels”. In: *Proceedings of the 1995 ACM SIGMOD International Conference on Management of Data*. SIGMOD ’95. New York, NY, USA: Association for Computing Machinery, pp. 1–10. ISBN: 0897917316. DOI: 10.1145/223784.223785. URL: <https://doi.org/10.1145/223784.223785>.
-  Bouajjani, Ahmed, Constantin Enea, and Enrique Román-Calvo (2025). “On the Complexity of Checking Mixed Isolation Levels for SQL Transactions”. In: *Computer Aided Verification - 37th International Conference, CAV 2025, Zagreb, Croatia, July 23-25, 2025, Proceedings, Part IV*. Ed. by Ruzica Piskac and Zvonimir Rakamaric. Vol. 15934, pp. 315–337. URL: https://doi.org/10.1007/978-3-031-98685-7%5C_15.
-  Cerone, Andrea, Giovanni Bernardi, and Alexey Gotsman (2015). “A Framework for Transactional Consistency Models with Atomic Visibility”. In: *26th International Conference on Concurrency Theory (CONCUR 2015)*. Vol. 42. Leibniz International Proceedings in Informatics (LIPIcs). Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, pp. 58–71. DOI: 10.4230/LIPIcs.CONCUR.2015.58.

References II

-  Cerone, Andrea and Alexey Gotsman (Jan. 2018). “Analysing Snapshot Isolation”. In: *J. ACM* 65.2. DOI: 10.1145/3152396. URL: <https://doi.org/10.1145/3152396>.
-  “Generalized Isolation Level Definitions” (2000). In: *Proceedings of the 16th International Conference on Data Engineering*. ICDE '00. USA: IEEE Computer Society, p. 67. ISBN: 0769505066.
-  Sun, Weihua and Zhaonian Zou (2025). *Vbox: Efficient Black-Box Serializability Verification*. arXiv: 2503.05163 [cs.DB]. URL: <https://arxiv.org/abs/2503.05163>.

Predicate results are observations over a covered set.



A principled basis for checking and analyzing databases with predicates.